9A Lesson 2

Inherited variation

Year 9 Science · 9A Genetics and evolution · Student Book pages 8 to 9

Name _____ Class _____ Date _____

Today I can:

- ☐ I can give examples of inherited variation and explain how sexual reproduction causes it.
- ☐ I can sort inherited variation into continuous and discontinuous, and recognise a normal distribution.

Key words

inherited variation differences in characteristics that are passed on from parents to offspring
offspring the young produced by parents
genetic information the instructions for inherited characteristics, stored in the nucleus
gene a section of DNA that carries the instructions for one characteristic
nucleus the part of a cell that contains the genetic information
gamete a sex cell (a sperm cell or an egg cell)
fertilisation when a male gamete and a female gamete fuse (join together)
zygote the fertilised egg cell
continuous variation variation that can have any value between two points
discontinuous variation variation with only a limited set of possible values
normal distribution the bell shape you get when grouped continuous data are plotted

● Core Everyone does this section.

1. Complete the sentences using words from the box. Use each word once.

Word box: characteristics fertilisation fuse genetic inherited offspring parents zygote

The nucleus of a cell contains the _____ information that controls _____ characteristics.

When two gametes _____ (join together) during _____, they produce a _____ (fertilised egg cell).

The fertilised egg cell contains genetic information from both _____, so it has instructions for inherited _____ from both parents. The _____ will have some characteristics of each parent.

2. Tick one box in each row to show what causes the variation: genes, the environment, or both.

Characteristic	Genes	Environment	Both
being able to speak more than one language			
length of hair			
human height			
natural eye colour			
blood group			
a scar			

3. Class survey. Record the class totals in the table.

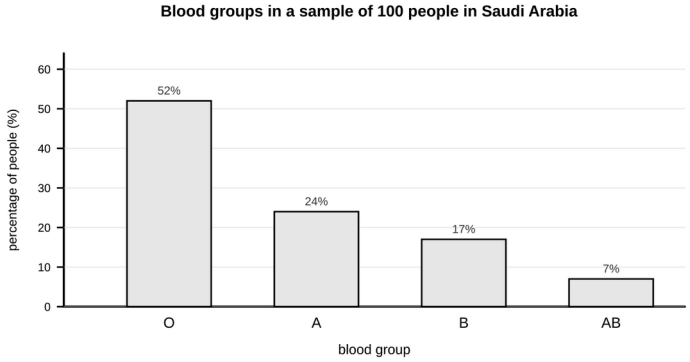
a) Complete the table using the class results.

Characteristic	Group	Number of students
tongue rolling	can roll	
	cannot roll	
ear lobes	attached	
	free (hanging)	

b) Is tongue rolling continuous or discontinuous variation? Give a reason.

●● Stretch When Core is finished.

S1. The bar chart shows the blood groups of people in Saudi Arabia.



a) What percentage of people have blood group A?

b) In a group of 300 people from Saudi Arabia, how many would you expect to have blood group AB? Show your working.

c) Tick the boxes that describe this type of variation.

- ☐ continuous
- ☐ discontinuous
- ☐ environmental
- ☐ inherited

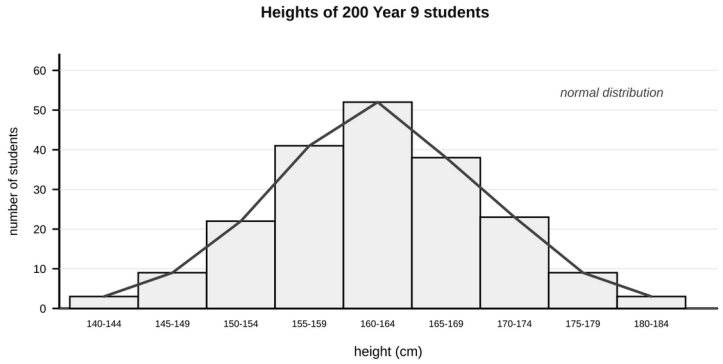
S2. Explain why foot length is an example of continuous variation but shoe size is discontinuous.

S3. Explain why brothers and sisters do not look exactly the same, even though they have the same parents.

●●● Challenge Explain in full sentences.

C1. Suggest a variation in tomatoes that is caused by both genetic information and environmental factors. Explain your reasoning.

C2. The chart shows the heights of 200 Year 9 students.



a) Describe the shape of the chart and name this type of distribution.

b) Explain why height gives this shape but blood group cannot.

c) Suggest one other characteristic that you would expect to show a normal distribution.

My thinking (any language)

Identical twins have exactly the same genetic information. Suggest how two identical twins could still look a little different from each other.

Word bank inherited variation · offspring · genetic information · gene · nucleus · gamete · fertilisation · zygote · continuous variation · discontinuous variation

✂ ---------- cut here

Exit ticket 9A Lesson 2: Inherited variation Name _____

1. Where in a cell is genetic information stored?

_____

2. Name the cell formed when two gametes fuse.

_____

3. Blood group: is it continuous or discontinuous, and inherited or environmental?

_____

Today I can: Goal 1 ☐ Goal 2 ☐